TESTS OF INDEPENDENCE USING CONTINGENCY TABLE DATA

Learning Outcome

- explain tests of independence based on contingency table data

When faced with categorical or discrete data, we cannot use the methods that we have discussed up to this point to test whether the classifications of such data are independent. Suppose we observe the following **frequency table** of 1,594 exchange-traded funds (ETFs) based on two classifications: size (i.e., market capitalization) and investment type (value, growth, or blend), as shown in Exhibit 7. The classification of the investment type is discrete, so we cannot use correlation to assess the relationship between size and investment type.

Exhibit 7: Size and Investment Type Classifications of 1,594 ETFs

	Size Based on Market Capitalization			
Investment Type	**Small**	**Medium**	**Large**	**Total**
Value	50	110	343	503
Growth	42	122	202	366
Blend	56	149	520	725
Total	148	381	1,065	1,594

Exhibit 7 is called a **contingency table** or a **two-way table** (because there are two classifications, or classes—size and investment type).

If we want to test whether a relationship exists between the size and investment type, we can perform a test of independence using a nonparametric test statistic that is chi-square distributed:

$$\chi^2 = \sum_{i=1}^{m} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad 4$$

where:

m = the number of cells in the table, which is the number of groups in the first class multiplied by the number of groups in the second class;

O_{ij} = the number of observations in each cell of row i and column j (i.e., observed frequency); and

E_{ij} = the expected number of observations in each cell of row i and column j, assuming independence (i.e., expected frequency).

This test statistic has $(r - 1)(c - 1)$ degrees of freedom, where r is the number of rows and c is the number of columns.

In Exhibit 7, size class has three groups (small, medium, and large) and investment type class has three groups (value, growth, and blend), so m is 9 (= 3 × 3). The number of ETFs in each cell (O_{ij}), the observed frequency, is given, so to calculate the chi-square test statistic, we need to estimate E_{ij}, the expected frequency, which is the number of ETFs we would expect to be in each cell if size and investment type are completely independent. The expected number of ETFs (E_{ij}) is calculated using the following:

$$E_{ij} = \frac{(\text{Total row } i) \times (\text{Total column } j)}{\text{Overall total}}. \qquad 5$$

Consider one combination of size and investment type, small-cap value:

$$E_{ij} = \frac{503 \times 148}{1,594} = 46.703.$$

We repeat this calculation for each combination of size and investment type (i.e., m = 9 pairs) to arrive at the expected frequencies, shown in Panel A of Exhibit 8.

Next, we calculate

$$\frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

the squared difference between observed and expected frequencies scaled by expected frequency, for each cell as shown in Panel B of Exhibit 8. Finally, by summing the values of

$$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

for each of the m cells, we calculate the chi-square statistic as 32.08025.

A. Expected Frequency of ETFs by Size and Investment Type			
	Size Based on Market Capitalization		
Investment Type	**Small**	**Medium**	**Large**
Value	46.703	120.228	336.070
Growth	33.982	87.482	244.536
Blend	67.315	173.290	484.395
Total	148.000	381.000	1,065.000

B. Scaled Squared Deviation for Each Combination of Size and Investment Type			
	Size Based on Market Capitalization		
Investment Type	**Small**	**Medium**	**Large**
Value	0.233	0.870	0.143
Growth	1.892	13.620	7.399
Blend	1.902	3.405	2.617

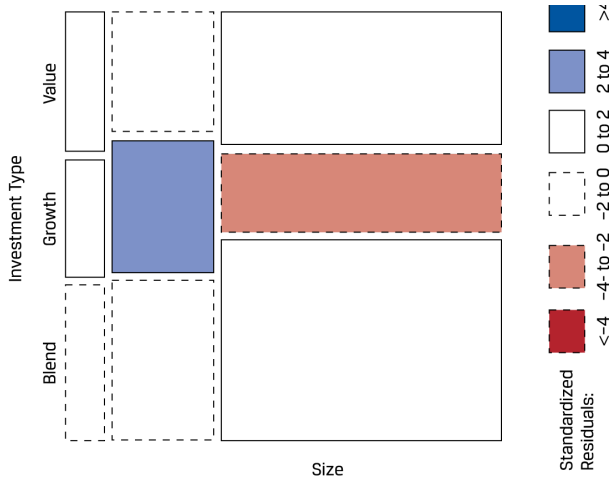
In our ETF example, we test the null hypothesis of independence between the two classes (i.e., no relationship between size and investment type) versus the alternative hypothesis of dependence (i.e., a relationship between size and investment type) using a 5 percent level of significance, as shown in Exhibit 9. If the observed values are equal to the expected values, the calculated test statistic would be zero. If, however, the observed and expected values are different, these differences are squared, so the calculated chi-square statistic will be positive. Therefore, for the test of independence using a contingency table, there is only one rejection region, on the right side.

Exhibit 9: Test of Independence of Size and Investment Type for 1,594 ETFs

Step 1	State the hypotheses.	H_0: ETF size and investment type are not related, so these classifications are independent; H_a : ETF size and investment type are related, so these classifications are not independent.
Step 2	Identify the appropriate test statistic.	$\chi^2 = \sum_{i=1}^{m} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
Step 3	Specify the level of significance.	5%
Step 4	State the decision rule.	With (3 − 1) × (3 − 1) = 4 degrees of freedom and a one-sided test with a 5% level of significance, the critical value is 9.4877. We reject the null hypothesis if the calculated χ^2 statistic is greater than 9.4877. **Excel** `CHISQ.INV(0.95,4)` **R** `QCHISQ(.95,4)` **Python** `FROM SCIPY.STATS IMPORT CHI2` `CHI2.PPF(.95,4)`
Step 5	Calculate the test statistic.	χ^2 = 32.08025
Step 6	Make a decision.	Reject the null hypothesis of independence because the calculated χ^2 test statistic is greater than 9.4877. Evidence is sufficient to conclude that ETF size and investment type are related (i.e., not independent).

We can visualize the contingency table in a graphic referred to as a mosaic. In a mosaic, a grid reflects the comparison between the observed and expected frequencies. Consider Exhibit 10, which represents the ETF contingency table.

Exhibit 10: Mosaic of the ETF Contingency Table



The width of the rectangles in Exhibit 10 reflect the proportion of ETFs that are small, medium, and large, whereas the height reflects the proportion that are value, growth, and blend. The darker shading indicates whether the number of observations is more than expected under the null hypothesis of independence, whereas the lighter shading indicates that the number of observations is less than expected, with "more than" and "less than" determined by reference to the standardized residual boxes. The standardized residual, also referred to as a Pearson residual, is as follows:

$$\text{Standardized residual} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}. \qquad 6$$

The interpretation for this ETF example is that there are more medium-size growth ETFs (standardized residual of 3.69) and fewer large-size growth ETFs (standardized residual of −2.72) than would be expected if size and investment type were independent.

EXAMPLE 4

Using Contingency Tables to Test for Independence

Consider the contingency table in Exhibit 11, which classifies 500 randomly selected companies on the basis of two environmental, social, and governance (ESG) rating dimensions: environmental rating and governance rating.

Exhibit 11: Classification of 500 Randomly Selected Companies Based on Environmental and Governance Ratings

	Governance Rating			
Environmental Rating	**Progressive**	**Average**	**Poor**	**Total**
Progressive	35	40	5	80
Average	80	130	50	260
Poor	40	60	60	160
Total	155	230	115	500

1. What are the expected frequencies for these two ESG rating dimensions if these categories are independent?

Hide Solution

Solution:

The expected frequencies based on independence of the governance rating and the environmental rating are shown in Panel A of Exhibit 12. For example, using Equation 5, the expected frequency for the combination of progressive governance and progressive environmental ratings is

$$E_{ij} = \frac{155 \times 80}{500} = 24.80.$$

The scaled squared deviations for each combination of environmental and governance rating are shown in Panel B of Exhibit 12. For example, using Equation 4, the scaled squared deviation for the combination of progressive governance and progressive environmental ratings is as follows:

$$= \frac{(35 - 24.8)^2}{24.8} = 4.195.$$

Governance Ratings

A. Expected Frequencies of Environmental and Governance Ratings Assuming Independence

	Governance Rating		
Environmental Rating	**Progressive**	**Average**	**Poor**
Progressive	24.8	36.8	18.4
Average	80.6	119.6	59.8
Poor	49.6	73.6	36.8

B. Scaled Squared Deviation for Each Combination of Environmental and Governance Ratings

	Governance Rating		
Environmental Rating	**Progressive**	**Average**	**Poor**
Progressive	4.195	0.278	9.759
Average	0.004	0.904	1.606
Poor	1.858	2.513	14.626

2. Using a 5 percent level of significance, determine whether these two ESG rating dimensions are independent of one another.

Hide Solution

Solution:

Step 1	State the hypotheses.	H_0: Governance and environmental ratings are not related, so these ratings are independent; H_a: Governance and environmental ratings are related, so these ratings are not independent.
Step 2	Identify the appropriate test statistic.	$\chi^2 = \sum_{i=1}^{m} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
Step 3	Specify the level of significance.	5%

Step 4	State the decision rule.	With (3 − 1) × (3 − 1) = 4 degrees of freedom and a one-sided test with a 5% level of significance, the critical value is 9.487729.
		We reject the null hypothesis if the calculated χ^2 statistic is greater than 9.487729.
		Excel
		`CHISQ.INV(0.95,4)`
		R
		`QCHISQ(.95,4)`
		Python
		`FROM SCIPY.STATS IMPORT CHI2`
		`CHI2.PPF(.95,4)`

Step 5	Calculate the test statistic.	χ^2= 35.74415 To calculate the test statistic, we first calculate the squared difference between observed and expected frequencies scaled by expected frequency for each cell, as shown in Panel B of Exhibit 12. Then, summing the values in each of the *m* cells (see Equation 4), we calculate the chi-square statistic as 35.74415.
Step 6	Make a decision.	Reject the null hypothesis because the calculated χ^2 test statistic is greater than 9.487729. Evidence is sufficient to indicate that the environmental and governance ratings are related, so they are not independent.

Question Set

An analyst group follows 250 firms and classifies them in two dimensions. First, they use dividend payment history and earnings forecasts to classify firms into one of three groups, with 1 indicating the dividend stars and 3 the dividend laggards. Second, they classify firms on the basis of financial leverage, using debt ratios, debt features, and corporate governance to classify the firms into three groups, with 1 indicating the least risky firms based on financial leverage and 3 indicating the riskiest. The classification of the 250 firms is as follows:

Financial Leverage Group	1	2	3
1	40	40	40
2	30	10	20
3	10	50	10

Using the classification of the 250 firms, answer the following questions:

1. What are the null and alternative hypotheses to test whether the dividend and financial leverage groups are independent of one another?

Hide Solution

Solution:

The hypotheses are as follows:

H_0: Dividend and financial leverage ratings are not related, so these groupings are independent.

H_a: Dividend and financial leverage ratings are related, so these groupings are not independent.

2. What is the appropriate test statistic to use in this type of test?

Hide Solution

Solution:

The appropriate test statistic is

$$\chi^2 = \sum_{i=1}^{m} \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

where O_{ij} represents the observed frequency for the *i* and *j* group and E_{ij} represents the expected frequency for the *i* and *j* group if the groupings are independent.

The expected frequencies based on independence are as follows:

	Dividend Group			
Financial Leverage Group	**1**	**2**	**3**	**Sum**
1	38.4	48	33.6	120
2	19.2	24	16.8	60
3	22.4	28	19.6	70
Sum	80	100	70	250

The scaled squared deviations for each combination of financial leverage and dividend grouping are:

	Dividend Group		
Financial Leverage Group	**1**	**2**	**3**
1	0.06667	1.33333	1.21905
2	6.07500	8.16667	0.60952
3	6.86429	17.28571	4.70204

3. If the critical value for the 0.05 level of significance is 9.4877, what is your conclusion?

Hide Solution

Solution:

The sum of these scaled squared deviations is the calculated chi-square statistic of 46.3223. Because this calculated value exceeds the critical value of 9.4877, we reject the null hypothesis that these groupings are independent.

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